Paper 1: Midterm Question Paper

RV University - School of Computer Science and Engineering 1111Course: Exploring Science - 1 (CS1804) 22Program: B.Tech (Hons.) 3Exam: CP2: Midterm Question Paper (Set 2) 4444

Date: Dec 06, 2024 | Time: 2:15 PM to 4:00 PM 5555

Max Marks: 25 | Semester: 1 6

Instructions:

- All questions are mandatory. Part A contains 5 questions of 1 mark each, and Part B contains 4 questions of 5 marks each. [7]
- Feel free to use calculators (simple or scientific) where necessary. Graphing calculators are not allowed. [8]
- Electronic devices (mobile phones, laptops, headphones/earpods, smart devices, etc.) are strictly prohibited. Using them may result in disqualification. [9]
- Present answers in a clear and structured manner. [10]
- Constants are provided at the end of the paper. [11]

Part A (1 Mark Each)

1. Identify correctly the number of nearest neighbors in a face-centered cubic structure? 12

a) 4

b) 6

c) 8

d) 12

2. Select the property of nanomaterials that explains their enhanced catalytic behavior. 13

a) Decreased density

b) Increased electrical conductivity

c) High surface area and reactivity

d) Improved crystalline structure

3. In Young's double-slit experiment, when both slits are open, an interference pattern is observed on the screen. This is because of superposition between the two states, namely, electron passing through the upper slit and the electron passing through the lower slit.

Evaluate and reason the impact of a measurement conducted to ascertain the slit through which the particle passes on the interference pattern. 14

a) The interference pattern remains unchanged because the particle's state is still in a superposition.

b) The interference pattern disappears because the act of measurement collapses the state and destroys the superposition.

c) The interference pattern becomes more distinct because the particle's state collapses, revealing which slit it passes through.

d) The interference pattern is unaffected because the measurement does not alter the state of the particle.

4. Evaluate the following statement: "The properties of nanomaterials are independent of their size." Is this statement: 15

a) True

b) False

5. Photolithography involves the use of light to: 16

a) Create high-energy electron beams for material ablation

b) Pattern photoresist layers on a substrate

c) Directly deposit atoms on a surface

d) Increase the grain size of nanoparticles

Part B (5 Marks Each)

6. A cubic crystal has planes with Miller indices (211) and a lattice parameter of 5 Å. 17

a) Calculate the interplanar spacing.

b) Determine the angle for second-order diffraction using X-rays of wavelength 9.89 Å.

7. 18

a) List any two synthesis techniques you will employ to obtain a nanomaterial in a laboratory. Explain each of the techniques in one or two sentences.

b) List any three crystal structures that have more than one Bravais lattice associated with them. Also, identify the Bravais lattices associated with them.

8. Consider a hot star with a surface temperature of 6657 K. 19

a) Determine the peak wavelength of its emitted radiation using Wien's displacement law.

b) Identify the hydrogenic transition responsible for this radiation using Rydberg's formula.

c) Classify the identified transition into its corresponding series.

9. 20

a) Explain why Bragg's law fails to detect amorphous materials but works effectively for crystalline solids.

b) In a Young's double-slit experiment, the distance between the two slits is 0.2 mm, and the screen is placed 1.2 m away. If the fourth-order bright fringe appears at 2.4 cm from the central maximum, calculate the wavelength of the light.

Constants: [21]

- $b = 2.89 \times 10^{-3} m \cdot K$
- $c = 3 \times 10^8 m \cdot s^{-1}$ (corrected from source typo)
- $R_H = 1.097 \times 10^7 m^{-1}$
- $k_B = 1.38 \times 10^{-23} J \cdot K^{-1}$

Paper 2: Degree Examination

RV University - School of Computer Science and Engineering 22222222Course: Exploring Science 1 (CS1804) 23232323Exam: B.Tech (Hons.) Degree Examination 24Semester: 1 25

Duration: 2 hours | Max. Marks: 30 26262626

Instructions:

- All questions are mandatory. Part A contains 2 questions (split into sub-questions) of 5 marks each, and Part B contains 2 questions of 10 marks each. [27]
- Feel free to use calculators (simple or scientific). Graphing calculators are not allowed. [28]
- Electronic devices are strictly prohibited. [29]
- All subsections of a question should be answered together. [30]

Part A

1. 31

a) Calculate the surface area to volume ratio and its error for a spherical nanomaterial with a radius of $10 \pm 0.1 \text{ nm}$.

b) The expression for Wien's displacement law is given as $\lambda T = b$. Modelling the star as a blackbody, calculate its surface temperature if the peak wavelength emitted is due to a Hydrogenic transition of wavelength 94.9 nm.

c) Explain the difference between a primitive unit cell and a unit cell.

d) How does the central dogma explain the flow of genetic information in a cell?

e) Why do mutations in DNA sometimes lead to diseases?

2. Describe the structure and function of lipids in cells. 32

Part B

3. 33

a) Explain how the Miller indices are related to a family of planes and thus, the inter-planar spacing.

b) State and explain Einstein's photoelectric equation. How does this photoelectric effect relate to the wave-particle duality of light and how does it challenge classical wave theories of light?

c) List and explain the five key experimental observations that classical theories of light could not explain, leading to the development of quantum theory.

4. 34

a) Compare and contrast the properties of bulk materials and nanomaterials in terms of their surface-to-volume ratio. Analyze how these differences influence the applications of nanomaterials in fields such as catalysis, drug delivery, and sensors.

b) Consider a compound that crystallizes in a body-centered cubic (BCC) Bravais lattice. Identify the number of lattice points, n, associated with the unit cell. Using this information, calculate what fraction of the unit cell volume is occupied by atoms of this compound, if its atomic radius is 0.128 nm.

c) A cubic compound with lattice constant 1 Å undergoes X-ray diffraction. The most intense peak of the diffraction pattern was found to be consisting of diffraction due to at least two different family of planes. The diffraction due to the first family of planes results in first order of diffraction $(n=1)$. The diffraction due to second family of planes results in second order of diffraction $(n=2)$. If the Miller indices of the first family of planes was (200), obtain the Miller indices of all the planes involved in forming the diffraction peak.

Hint: The wavelength λ and the glancing angle θ remain the same, implying that $d_1/n_1 = d_2/n_2$. Making use of this relation, obtain the Miller indices.

Constants: 35

- $b = 2.89 \times 10^{-3} m \cdot K$
- $c = 3 \times 10^8 m \cdot s^{-1}$
- $R_H = 1.097 \times 10^7 m^{-1}$
- $k_B = 1.38 \times 10^{-23} J \cdot K^{-1}$
- $m_e = 9.1 \times 10^{-31} kg$
- $q = 1.6 \times 10^{-19} C$
- $h = 6.626 \times 10^{-34} J \cdot s$
- $g = 9.8 m \cdot s^{-2}$